DATA COLLECTION SYSTEM



DRUMS

DAIKAI REMOTE UBIQUITOUS MONITORING SYSTEM

3rd March 2026

Version 7

Daikai Engineering Pte Ltd

WHAT IS DRUMS?



Data Collection & Monitoring



Machine Learning & Dashboards



Actionable Insights



WHY DRUMS?



Centralised Data



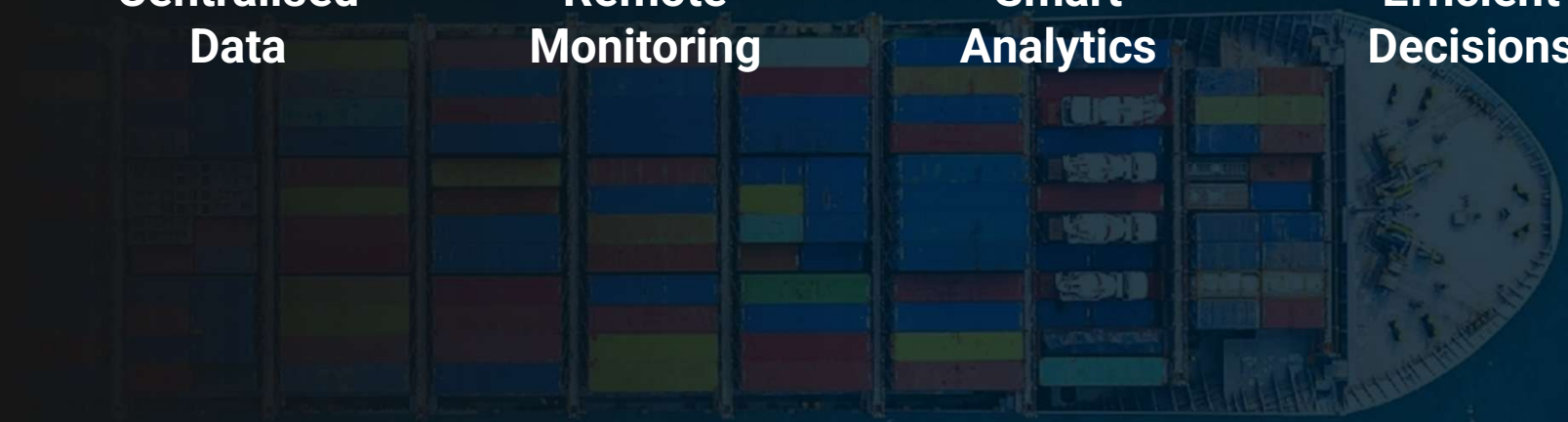
Remote Monitoring



Smart Analytics



Efficient Decisions





TECHNOLOGY ADVANTAGES & LIMITATIONS



Near Real-Time Monitoring



Centralised Data Platform



Operational Efficiency



Predictive Maintenance

➔ Considerations

- Requires stable connectivity and initial deployment cost

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DATA SECURITY & SYSTEM PROTECTION



Secure Data Transmission



Authentication & Access Control



Multi-Layer Security



Vessel-Level Data Protection



Audit & Monitoring

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DRUMS CONFIGURATION

- Collection of data from Daihatsu DECSS via Serial Interface

MODBUS, DECSS, individual wiring

Internet

Internet

Sensors



Data Collector



Remote Monitoring Server



DRUMS website



Approx.
11.2 KB CSV file

Compressed & encrypted

Data of ~51.2 KB is instantly transmitted every 3 minutes.

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CONNECTIVITY OPTIONS



Wi-Fi

Wi-Fi router

RJ45/ethernet

Router

Cellular network

Cellular dongle

The Vessel internet

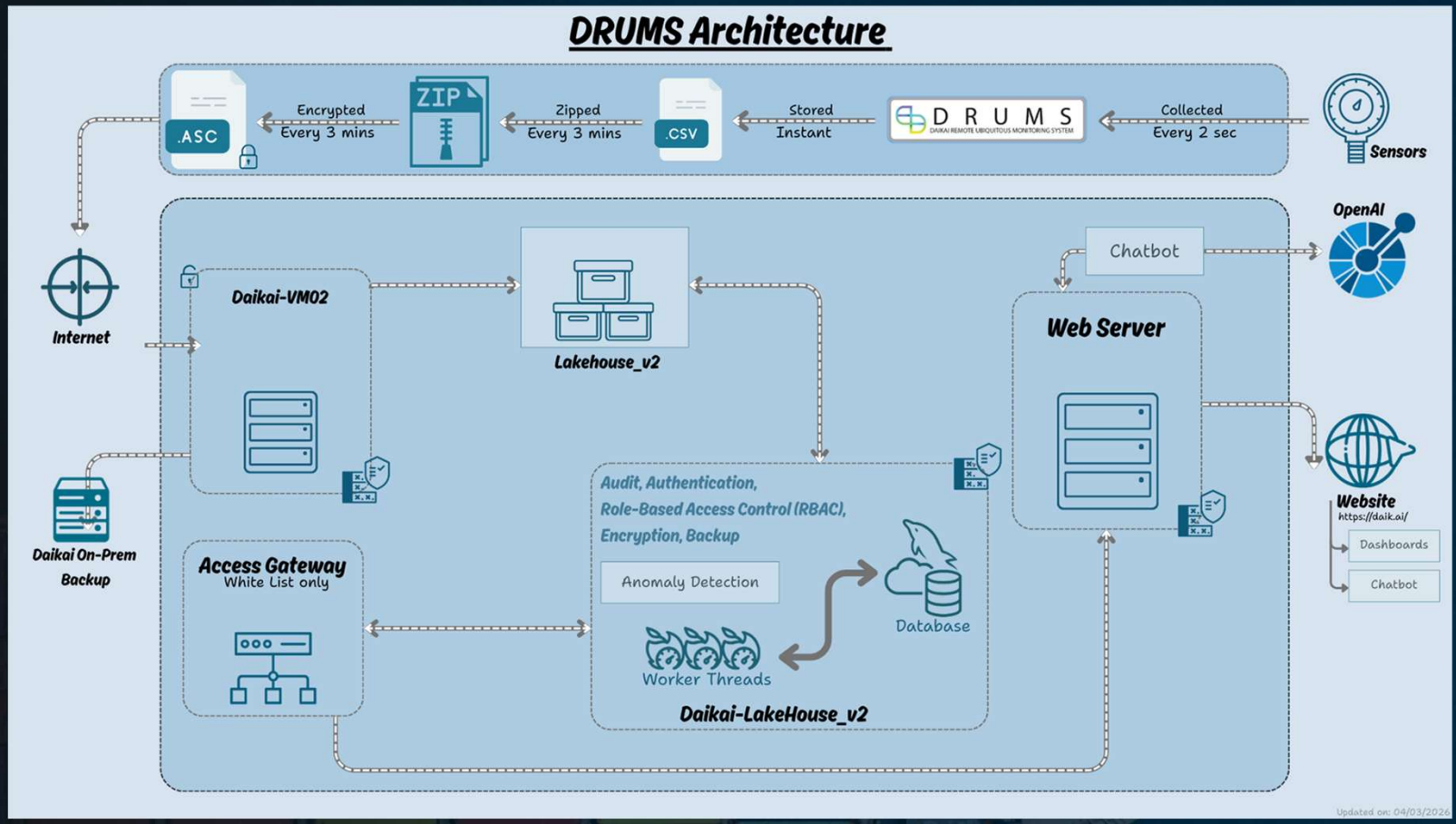
Cloud Server

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DRUMS ARCHITECTURE



SUPPLY RECORDS

Vitol (4 Vessels)

Emarat (6 Vessels)

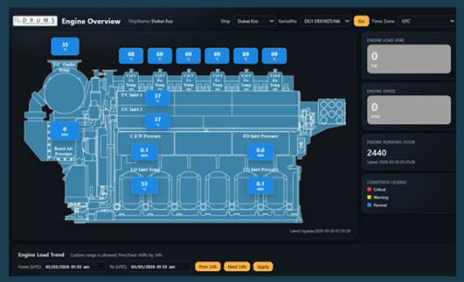
Truongminh (4 Vessels)

Jemla (1 Vessel)

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CORE FEATURES



Dashboards



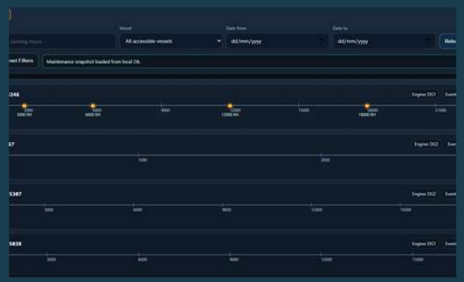
AI Chatbot



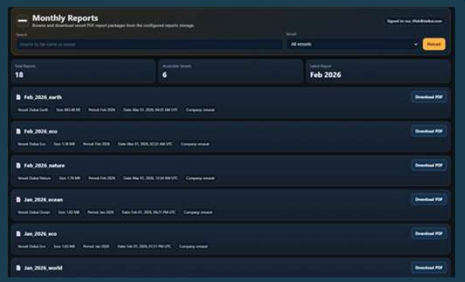
Homepage Console



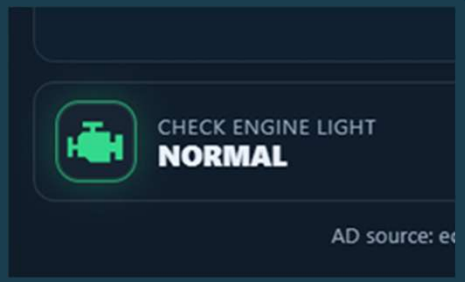
Alarms



Maintenance Record



Monthly Report



Anomaly Detection



Exhaust

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